Design of Experiments as a Preprocessing Tool for Model Development in Two Examples

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Abstract

TBD

Introduction

One of the foundation tools devised for statistical experimental design and general data science was design of experiments (DOE). The idea and original work was by R. A Fisher. His book, *Statistical Methods for Research Workers*, on the subject was published in 1925. Fisher's 1925 book was clearly formative, but his *The Design of Experiments* (1935) is the more direct foundational DOE reference. The landmark work in 1951 by George E. P. Box and K. B. Wilson expanded this foundation. Taken all together, concepts such as replication, factorial designs, and analysis of variance were introduced and connected in useful ways for planning experiments and analyzing their data.

Back in the 1920s and 1930s experiments data was often sparse and slow and expensive to generate, and the invention of transistors and the dawn of the digital age were decades away. Generations of experimental engineers and scientists needed reliable and repeatable ways to distinguish important inputs and their effects on measured outputs. And over time the science of DOE was proven to fill that need. As the computers and the digital age developed in the 1950s and 1960s, other concepts and tools connected to DOE were devised; e.g., response surfaces and surrogate models, Monte Carlo simulation, probabilistic design.

In this age of machine learning and AI, the role of DOE is sometimes underemphasized in workflows dominated by predictive modeling. That's unfortunate because at a minimum it's still a very useful preprocessing tool for or alongside model development. As a preprocessor it offers a well-understood, structured, statistical method for quickly studying cause-and-effect relations between input variables and corresponding output responses. Used property, DOE can help engineers identify promising operating setting for subsequent machine learning analyses.

By means of 2 examples, this report explains how DOE can be used as a preprocessor. These examples are small in terms of their I/O. The focus here is to demonstrate the usefulness of DOE for modest problems. In that connection it's important to mention that the DOE approach becomes unwieldy when large number of inputs (> 100) are involved. This is because full factorial designs become unmanageable far earlier. For example, 2-level factorial analysis with 2 input factors requires only 4 combinations and 4 runs; and 20 factors implies 1,048,576 combinations. Higher dimensional applications may still use screening techniques (e.g.,, fractional factorial). Even with aggressive screening factors greater than 100 are difficult to process and often infeasible.

First Example: Terminal-Velocity Approach of a Body Falling Through Air

This example uses a simple nonlinear Newtonian-physics problem with a known analytical solution: the vertical fall of a body through still air under aerodynamic drag. The objective is to determine the time required for a body to reach a specified fraction of its terminal velocity,. As stated, the problem presents a nonlinear ordinary differential equation (ODE) that is solvable. Because the physical response is available, the problem provides a useful benchmark for evaluating the accuracy and practical value of a DOE-based surrogate model.

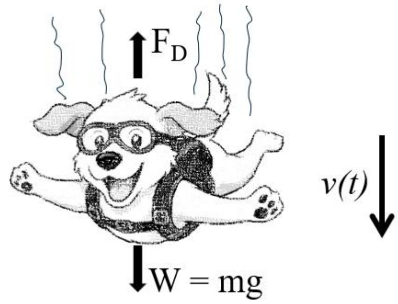

Figure 1 – A Free-Falling Body

Assume that the body falls vertically from rest through still air, that its orientation and projected frontal area remain constant, and that air density is constant. Let downward velocity be positive. The forces acting on the body are its weight, $W = mg$, acting downward, and drag, F_D, acting upward:

$$F_D = \frac{1}{2}\rho C_D A v^2 \quad (1)$$

where ρ (= 1.23 kg/m^3 at sea level) is air density, C_D is the drag coefficient, A is the projected frontal area normal to the direction of motion, and v is the body's speed relative to the air. The motion of a falling body can be described by Newton's 2nd law of motion:

$$ma = mg - F_d \quad (2)$$

Where a is the body's acceleration. Substituting (1) into the above and replacing a with the derivation of velocity, $a = dv/dt$, we have

$$m\frac{dv}{dt} = mg - \frac{1}{2}\rho v^2 C_D A v^2 \quad (3)$$

This is a first-order nonlinear ODE to be solved for the single, time-dependent unknown, $v(t)$, with the aim of determining the time it takes to its terminal velocity. At terminal velocity, the force of gravity perfectly balances the drag force, so than the net force and acceleration are zero. Adjusting (3) accordingly gives the terminal velocity:

$$v_T = \sqrt{\frac{2mg}{\rho C_D A}} \quad (4)$$

The body approaches v_T asymptotically and therefore does not attain it at a finite time. For engineering purposes, define t_{95} as the time required to reach 95% of terminal velocity:

$$v(t_{95}) = 0.95 v_T \quad (5)$$

Equation (3) can be solved mathematical or numerically. Results are easily obtained by using a little bit of calculus combined with a simple finite different approach to determine $v(t_{95})$, which is the response of interest.

The ODE (Eq. 3) presents 3 inputs: $C_D, A, \& \, m$, with ranges

$$\begin{aligned} X_1 &= C_D, \quad (0.5 \le C_D \le 1.0) \\ X_2 &= A, \quad (0.05\, m^2 \le A \le 0.5\, m^2) \\ X_3 &= m \quad (5\, kg \le m \le 65\, kg) \end{aligned} \tag{6}$$

The range of values for C_D represent 0.5 for a sphere and 1.0 for a human body. For A and m, the ranges represent physically plausible values for common household objects (e.g., a 5-kg cast iron skillet, 65-kg television).

With reference to Eq. 5 we have one output:

$$\mathrm{y} = t_{95} \tag{7}$$

Turning attention to the DOE analysis, the input factor values are defined at 2 levels (low and high), as shown in Table 1

Factor	High Value (+)	Low Value(+)
C_d	1	0.5
A	0.5	0.05
m	65	5

Table 1- Input levels

A full factorial design requires 2^3 = eight analyses. For each combination of low and high factor levels, the exact expression for is evaluated and given in Table 2. The resulting response values range from 2.4 to 38.0 seconds for the selected design space.

	Factors			Response
Analysis	$X_1 (C_D)$	$X_2 (A)$	$X_3 (m)$	$y(t_{95})$ (sec)
1	1	0.5	65	8.5
2	1	0.5	5	2.4
3	1	0.05	65	26.9
4	1	0.05	5	7.5
5	0.5	0.5	65	12.0
6	0.5	0.5	5	3.3
7	0.5	0.05	65	38.0
8	0.5	0.05	5	10.6

Table 2- Results for the Nonlinear ODE Analysis

For DOE analysis, the actual factors are transformed to normalized factors over the interval.

$$-1 \leq x_i \leq 1, \text{ for } \mathrm{i} = 1, 2, 3. \tag{8}$$

The coded first-order model including all two-factor interactions is

$$t_{95} = a_0 + a_1x_1 + a_2x_2 + a_3x_3 + b_{12}x_1x_2 + b_{13}x_1x_3 + b_{23}x_2x_3 \tag{10}$$

Main effect results and two-factor interaction results are given in Figure 3. This etc. Clearly factors m and A are the dominant factors. The regression's main and interaction coefficients are also computed and substituted into Eq. (10), which becomes

$$\begin{aligned} \mathrm{y}' = t^* &= 13.65 - 2.325x_1 - 7.100x_2 + 7.700x_3 + \cdots \\ &1.225x_2 - 1.325\ 6x_1x_3 - 4.000x_2x_3 \end{aligned} \tag{11}$$

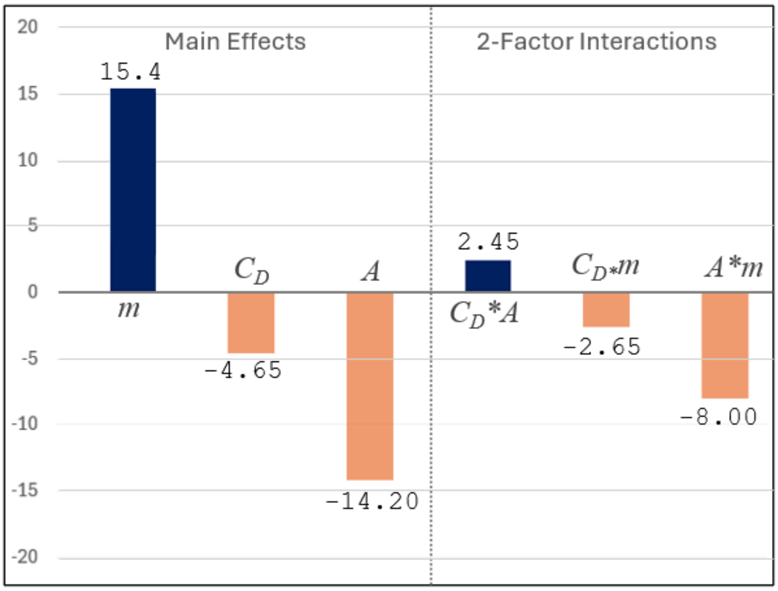

Figure 2 – DOE Results

As suggested earlier, this equation can be used as a DOE-based surrogate model. It's adjusted $R^2 = 0.973$.

Second Example: A Regression and DOE-Inspired Analysis of NFL Point Spreads"

Unlike the first example, this case study uses observational data from a real-world setting. NFL game outcomes are influenced by many potential predictors and exhibit substantial unexplained variation. Each observation represents one game between two teams, denoted Team A and Team B.

The objective is to model the final point differential

$$S_{AB} = P_A - P_B$$

where P_A and P_B are the final points scored by Teams A and B, respectively. Thus, $S_{AB} > 0$ indicates a Team A victory, whereas $S_{AB} < 0$ indicates a Team B victory.

Many pregame variables could be considered, including team win percentage, recent win–loss performance, previous-game scoring margin, home-field status, and market point spread. To demonstrate the use of regression and DOE-style factor analysis, the analysis is limited to six predictors and one response:

$$\begin{aligned}
X_1 &= \text{Team A's winning } (+) \text{ or losing } (-) \text{streak} \\
X_2 &= \text{Team B's winning } (+) \text{ or losing } (-) \text{streak} \\
X_3 &= \text{Team A's point differential in its previois game} \\
X_4 &= \text{Team B's point differential in its previois game} \\
X_5 &= 1 \text{ if Team A is the home team or} = -1 \text{ if Team A is the away team} \\
X_6 &= \text{The gambling point spread at game time}
\end{aligned}$$

The point spread is recorded from Team A's perspective. A negative value means Team A is favored; for example, $X_6 = -7$ means Team A is a 7-point favorite. A positive value means Team A is the underdog; for example, $X_6 = +7$ means Team A is a 7-point underdog. Team A covers the spread when

$$S_{AB} + X_6 > 0$$

If $S_{AB} + X_6 = 0$, the wager is a push for an integer point spread.

The dataset consists of games from the 2020 NFL regular season. Unlike a designed factorial experiment, these predictor values were not assigned at controlled high and low settings; they are naturally occurring, and the resulting data may be correlated and unbalanced. Consequently, the analysis is best viewed as a DOE-inspired regression analysis of observational data.

Before fitting the model, predictors X_1, X_2, X_3, X_4, and X_6 are rescaled to normalized variables on $[-1, +1]$. The home/away variable X_5 already lies on this interval. A first-order regression model with all two-factor interactions is then fitted:

$$\tilde{y} = a_0 + a_1x_1 + a_2x_2 + \cdots a_6x_6 + b_{12}x_1x_2 + b_{13}x_1x_3 + \cdots b_{56}x_5x_6$$

Here, $\tilde{y}$ is the predicted point differential for Team A relative to Team B. The model contains six main effects and 15 two-factor interactions, in addition to the intercept. Model performance should be assessed using chronological out-of-sample validation, because the objective is prediction of future games rather than explanation of games already observed.

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